

006 Minimum Formal / Mechanical Reference

A Page-Zero Deck for Typed Mathematical Use

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Status. DECLARED as a Language Mechanics reference instrument. Definitions are constitutive declarations of this deck; propositions and corollaries below are DERIVED from those declarations.

Abstract. This deck fixes the minimum typing required to carry a formal reference through an operation without silently exchanging reader equality, carrier equality, history, lift, or refusal. It isolates a reference packet, typed return, receipt adequacy, recognition, the admission gate, and the distinction between *minimal* and *minimum*. A finite mechanical witness is included only after the operations it uses are declared: the seal carrier is a join-semilattice and the residue carrier is a monoid. The deck is locally complete and imports no physical, semantic, numerical, or cross-domain identification.

1 Position

This is a reference deck, not a replacement foundation for mathematics. It asks one operational question:

When a formal expression carries a referent through an operation, what must be declared so that the result can be compared with its reference without type loss?

The deck begins from ordinary formal objects, operations, equality relations, and maps. It adds only explicit typing of reference, return, readable difference, lift, receipt, recognition, minimum, and refusal.

2 The minimum reference packet

Fix a jurisdiction

$$D = (\Sigma, R, \rho),$$

where Σ is the candidate class, R is the permitted re-posing relation, and $\rho : \Sigma \rightarrow Y$ is the declared reader.

Definition 2.1 (Reference packet). *A formal reference packet is*

$$\mathfrak{R} = (D, x_0, T, \rho),$$

where $x_0 \in \Sigma$ is the addressed referent and $T : \text{Dom}(T) \subseteq \Sigma \rightarrow \Sigma$ is a declared partial operation.

The elementary transport is

$$x_0 \xrightarrow{T} T(x_0),$$

and the first return question is

$$\rho(Tx_0) \stackrel{?}{=} \rho(x_0).$$

No statement about identity, memory, meaning, or mechanism follows until its own office is declared.

3 Holding-through and typed equality

Definition 3.1 (Reference survives operation). *The reference survives T at x_0 under reader ρ when*

$$\rho(Tx_0) = \rho(x_0).$$

If the full carrier is the reader, $\rho = \text{id}_\Sigma$, this reduces to exact carrier return $Tx_0 = x_0$.

Proposition 3.2 (Typed equality). *From*

$$\rho(Tx) = \rho(x)$$

one may conclude only that Tx and x lie in the same fiber of ρ . Carrier equality $Tx = x$ follows only when the declared hypotheses make that fiber a singleton for the compared pair—for example, when ρ is injective there.

Proof. Equality of the reads says $Tx \in \rho^{-1}(\rho(x))$. Without injectivity, a fiber may contain distinct points, so equality in Y does not force equality in Σ . If ρ is injective on the relevant domain, the implication follows by definition of injectivity. $\square$

Status. Proposition 3.2 is DERIVED.

4 Minimum office register

The following table is an entrance index. Any stronger use of a term requires the stronger structure to be declared where it is used.

Office	Exact minimum use	Boundary
Jurisdiction	$D = (\Sigma, R, \rho)$: where a claim is posed	does not supply content
Reference	addressed comparison baseline	does not explain why selected
Condition	entry requirement	does not guarantee continuation
Constraint	relation preserved by admitted continuation	distinct from condition
can	lawful continuation in the declared jurisdiction	gate verdict, not truth
$\emptyset$	refused / undefined continuation here	not scalar zero
Operation	typed map or partial map	no semantics supplied by the map alone
Difference	distinguishable departure under shared reference	requires a comparison frame
Return	equality at a declared reader	not automatically carrier reset
Lift	finer carrier/address above a quotient or read	requires an explicit map
Receipt	recoverable evidence of occurrence at promised resolution	not the state itself
Recognition	warranted present comparison under a declared reader	not unrestricted identity
Minimum	least admitted construction under a declared order	stronger than mere minimality
Fit	witnessed compatibility preserving the required relation	resemblance is insufficient
Function	operation lawfully carried through admitted Fit	does not create Fit

5 Exact return, reader return, and lift

Two predicates must remain separate:

$$\text{exact carrier return: } Tx = x,$$

reader return with retained carrier difference: $\rho(Tx) = \rho(x), \quad Tx \neq x.$

The second form is the minimum opening for a lift: the declared read closes while a finer carrier distinction remains.

Definition 5.1 (Lift datum). *Given a quotient or reader $p : \tilde{X} \rightarrow X$, a lift datum for a returned action consists of a lifted action $\tilde{T}$ and a witness*

$$p(\tilde{T}\tilde{x}) = p(\tilde{x}),$$

with, whenever nontriviality is claimed,

$$\tilde{T}\tilde{x} \neq \tilde{x}.$$

The quotient return and the lifted difference are separate reads of one declared construction.

6 Receipt and recognition

A return may erase information from its displayed state. Receipt therefore occupies a separate office.

Definition 6.1 (Receipt adequacy). *Let H be a history class and let $\sim_\delta$ be the equivalence relation induced by the promised history resolution δ , with quotient map*

$$\pi_\delta : H \rightarrow H/\sim_\delta.$$

Let Σ_δ be a receipt space and

$$\sigma_\delta : H \rightarrow \Sigma_\delta$$

a receipt map. The receipt is adequate at resolution δ when the promised quotient remains recoverable. A sufficient exact condition is a decoder

$$\text{Dec}_\delta : \text{im}(\sigma_\delta) \rightarrow H/\sim_\delta$$

satisfying

$$\boxed{\text{Dec}_\delta \circ \sigma_\delta = \pi_\delta.}$$

The two sides now have the same domain H ; the equation states exactly that the receipt retains every history distinction promised by the quotient.

Definition 6.2 (Recognition). *Recognition at reader ρ is a warranted present comparison*

$$\text{Rec}_\rho(x; x_0) = 1$$

only when the declared comparison is defined, the stated equality or bounded relation holds, and the evidence required by the claim is present.

Recognition returns only what its reader and receipt can carry.

7 The gate and refusal

For a candidate move T at x , let $C(x)$ be its entry condition and let $Q(x, Tx)$ be the continuation constraint. Define

$$\text{Adm}(T, x) \iff C(x) \wedge (T \text{ is typed at } x) \wedge Q(x, Tx),$$

and

$$\text{Gate}(T, x) = \begin{cases} \text{can}, & \text{Adm}(T, x), \\ \emptyset, & \text{otherwise.} \end{cases}$$

The namespace is locked:

$$\boxed{\emptyset \neq 0.}$$

A refused continuation is not a scalar, zero vector, empty history, false proposition, or absent physical object.

8 Minimal and minimum

The word *minimum* carries a stronger burden than *minimal*.

Definition 8.1 (Minimal and minimum admitted constructions). *Let $(\mathcal{A}, \preceq)$ be a declared partially ordered class of admitted constructions, with strict part $\prec$. A construction $m \in \mathcal{A}$ is minimal when*

$$\nexists c \in \mathcal{A} \text{ such that } c \prec m.$$

It is a minimum when

$$\boxed{m \preceq c \quad \text{for every } c \in \mathcal{A}.}$$

Proposition 8.2 (Minimum implies minimal). *Every minimum is minimal. A minimal element need not be a minimum.*

Proof. If $m \preceq c$ for every $c \in \mathcal{A}$, no $c \prec m$ can exist. Conversely, two incomparable minimal elements supply a class with minimal elements but no minimum. $\square$

Therefore every formal use of *minimum* must declare at least:

- the admitted candidate class;
- the admission conditions;
- the comparison/reduction order;
- the relations that must be preserved;
- the proof that the selected construction lies below every admitted competitor when a true minimum is claimed.

Without those declarations, “minimum” is not yet a formal result.

9 Mechanical seal/residue witness

A finite witness can separate cumulative reference from carried non-closing difference, but its operations must be typed before use.

Let $(S, \vee)$ be a finite join-semilattice with its induced order $\leq$. Let $(A, \oplus_A, 0_A)$ be a monoid. Define

$$\Gamma = S \times A.$$

For a declared turn k , choose $e_k \in S$ and $\epsilon_k \in A$, and define

$$U_k(\lambda, \eta) = (\lambda \vee e_k, \eta \oplus_A \epsilon_k).$$

Corollary 9.1 (Well-defined monotone seal update). *Each $U_k : \Gamma \rightarrow \Gamma$ is well defined, and its seal coordinate is monotone:*

$$\lambda \leq \lambda \vee e_k.$$

Proof. The join exists by the join-semilattice law, and $\oplus_A$ is closed on A by the monoid law. The defining property of a join makes $\lambda \vee e_k$ an upper bound of λ . $\square$

The coordinates have different jobs:

seal records admitted cumulative structure

residue carries non-closing difference.

This witness imposes no historical twelve-coordinate semantic map, spectral/color read, distinguished numerical constant, calendar, or physical realization.

10 Page-zero rule and boundary

A formal term must survive the operation under which it is reused.

Expanded:

reference $\rightarrow$ typed operation $\rightarrow$ return test $\rightarrow$ receipt $\rightarrow$ recognition $\rightarrow$ reuse.

This is a discipline of formal use. It does not claim that mathematics, physics, language, or cognition are one mechanism. It does not import certification from a neighboring paper or from repeated vocabulary. Any stronger construction must state its own carrier, operations, reader, preserved relation, witness, and falsifier.

Source record

The following sources are provenance only; no theorem above depends on an unquoted result from them.

- A. V. Coppa, *Rule & Reference: Page Zero*, 10 June 2026.
- A. V. Coppa, *Classical Reference: A Return Typing of Classical Mathematics*, working paper, 2026.
- A. V. Coppa, *Minimum Mechanical Reference: Seal / Residue Form*, working paper, 2026.